RNA Extraction Protocol

This document is a cleaned and properly formatted English version of the attached RNA extraction notes for use with the EURx Universal RNA Purification Kit.

Materials

- RNase decontamination solution (e.g. RNase Ex).
- EURx **Universal RNA Purification Kit**; store as indicated by the manufacturer.
- DNase/RNase-free microcentrifuge tubes; take tubes directly from the bag and avoid touching inside the bag.
- Absolute ethanol (96-100%).
- DTT or beta-mercaptoethanol (beta-ME).
- Ice.
- Filter pipette tips.
- Cold PBS.
- RNase-free water.

Preparation

- Clean the working surface with RNase decontamination solution.
- Wear appropriate laboratory clothing and minimize exposed skin.
- Keep samples and tubes on ice whenever possible during the procedure.
- Before use, add 10 uL DTT or beta-ME per 1 mL of RL lysis buffer and keep the buffer on ice.

Cell Collection

1. Remove the culture medium from the cells.
2. Add trypsin in a volume sufficient to cover the surface of the plate.
3. Collect the cells by adding culture medium.
4. Transfer the suspension to a new microcentrifuge tube.
5. Centrifuge at 500 x g for 5 minutes at room temperature.
6. Remove the supernatant.
7. Add 1 mL cold PBS.
8. Centrifuge at 500 x g for 5 minutes at 4 C.
9. Remove the supernatant and keep the cell pellet on ice.

storage of cell pellet at
-20°C possible

RNA Purification

1. Add 400 uL RL buffer to the cell pellet.
2. Pipette up and down several times to ensure complete cell disruption.
3. Transfer the lysate to a **homogenization spin column** placed in a 2 mL receiver tube.
4. Centrifuge at 11,000 x g for 2 minutes.
5. Add 250 uL ethanol (96-100%) to the **flow-through**.
6. Mix thoroughly by pipetting. Do **not** centrifuge at this stage.
7. Transfer the mixture to an **RNA binding spin column** placed in a 2 mL receiver tube.
8. Centrifuge for 1 minute at 11,000 x g.
9. Discard the flow-through.
10. Add 400 uL **DN1 wash buffer**.
11. Centrifuge for 1 minute at 11,000 x g.
12. Remove the spin column, decant the supernatant, and place the column back into the receiver tube while minimizing contact with the lid.
13. Add 600 uL **RBW wash buffer**.
14. Centrifuge for 1 minute at 11,000 x g.
15. Remove the spin column, decant the supernatant, and return the column to the receiver tube.
16. Add 300 uL **RBW wash buffer**.
17. Centrifuge for 2 minutes at 11,000 x g. + decant
18. Return the column to the receiver tube and centrifuge again for 1 minute at 11,000 x g to dry the membrane as completely as possible.

Elution

1. Place the spin column into a new pre-cooled 1.5-2 mL receiver tube.
2. Add 40 uL RNase-free water directly onto the column membrane, taking care not to touch the membrane with the pipette tip.
3. Centrifuge for 1 minute at 11,000 x g.
4. Remove the spin column.

Storage

- Store the RNA immediately at 2-8 C (preferred for short-term use) or at -20 C.
- Avoid repeated freeze-thaw cycles.
- Alternatively, proceed directly to cDNA synthesis.

Notes

- Maintain RNase-free conditions throughout the workflow.
- Use fresh consumables and avoid unnecessary handling of tube openings, lids, and column membranes.
- This protocol is a cleaned transcription of the attached working notes and should be cross-checked with the official EURx kit manual before routine use.

RNA calibration:
1. turn on NanoDrop
2. select RNA
3. Blank measurement w/ RNA-free water

RNA analysis:
1. wipe with water
2. put 1 ul on test area and softly close the lid
3. measure
-repeat for all samples-
4. at the end, transfer all data on USB stick

Expected values:
- RNA concentration: >10 ng/ul
- A260/A280: >1,8
- A260/A230: >1,5